

Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

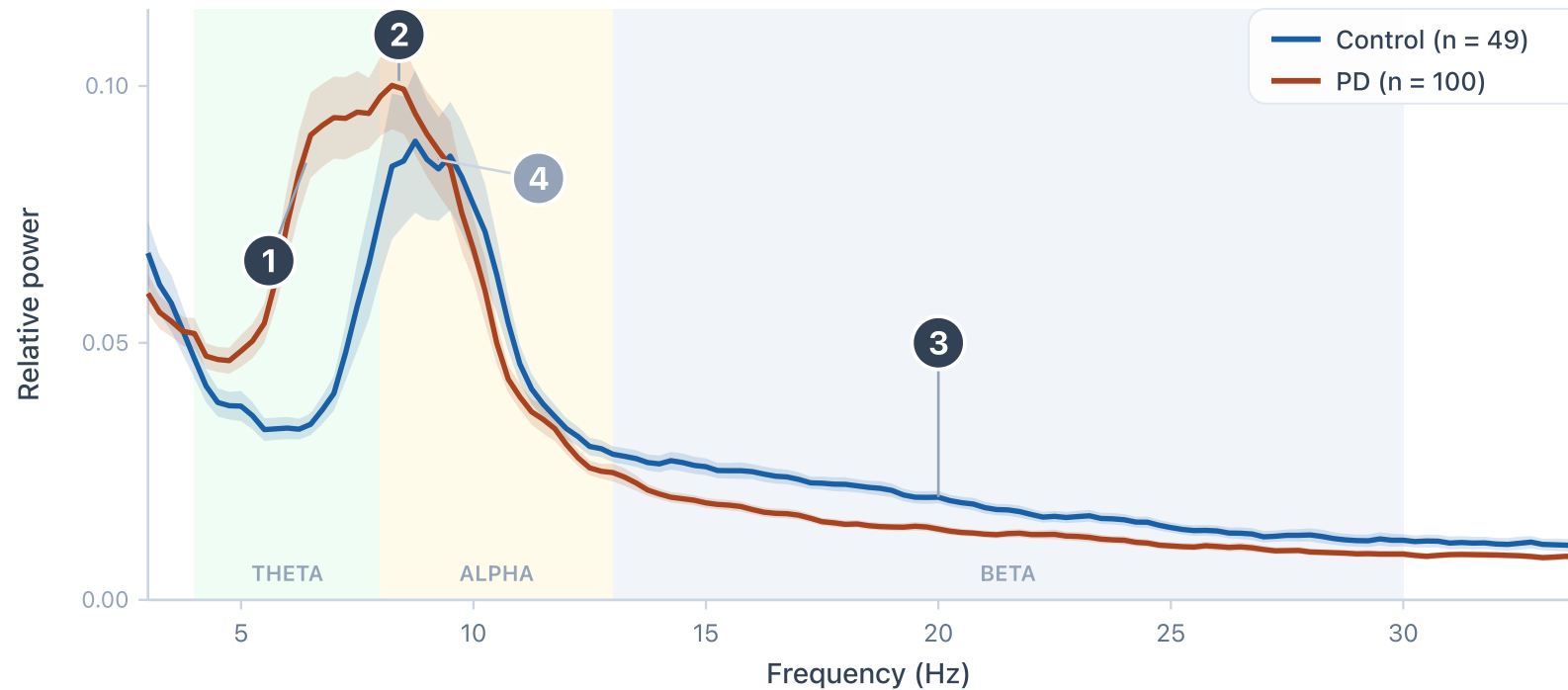
Hypotheses

① theta power higher
in PD · $p < 0.001$

② alpha peak slows
9.25 → 8.00 Hz · $p < 0.001$

③ beta power lower
in PD · $p < 0.001$

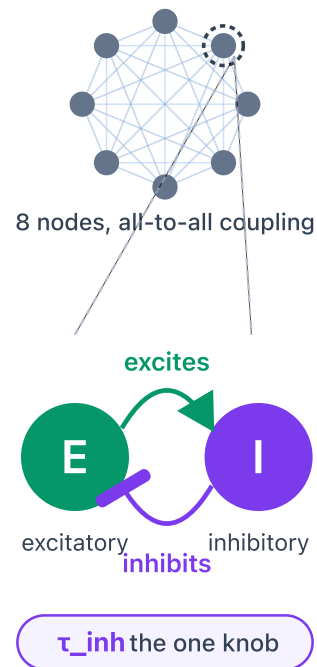
④ alpha power
unchanged · $p = 0.9$



Three changes confirmed, one prediction null: the slowing is in frequency, not amplitude.

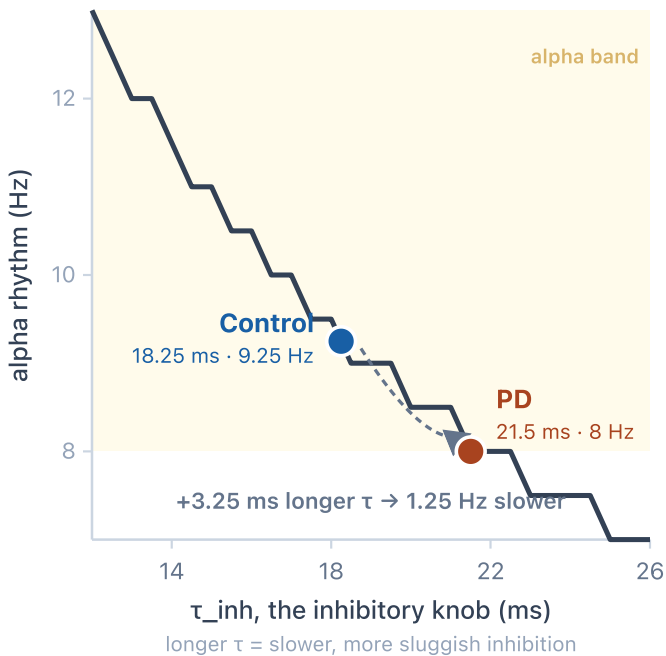
The model: a Wilson-Cowan network

THE MODEL: AN E-I LOOP

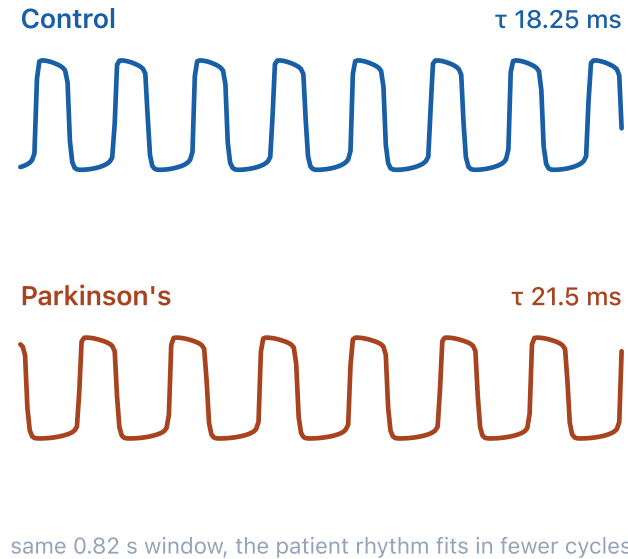


TURN τ_{inh} UP, THE RHYTHM SLOWS

each point = the model run at one knob setting



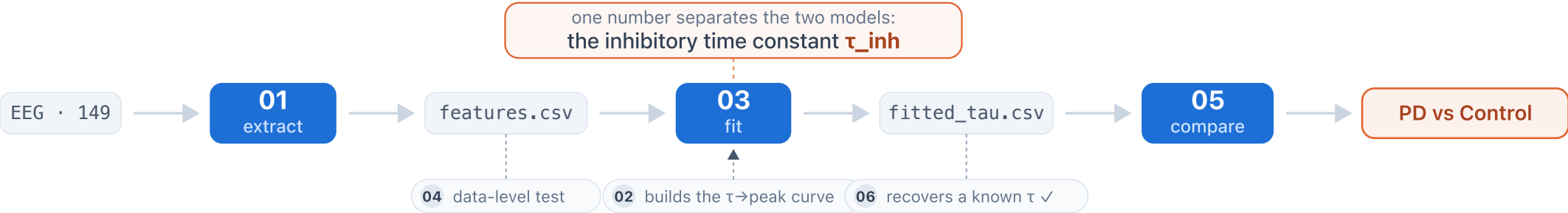
THE SIMULATED RHYTHM



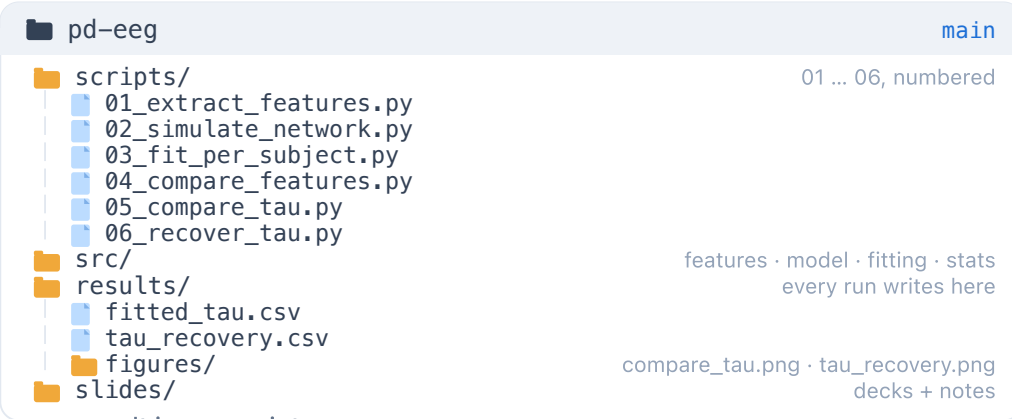
The whole change from healthy to patient is one number: a longer inhibitory time constant τ_{inh} , about **+3.25 ms** in PD.

Code & repository

THE PIPELINE EEG to result, through six numbered scripts



THE REPOSITORY

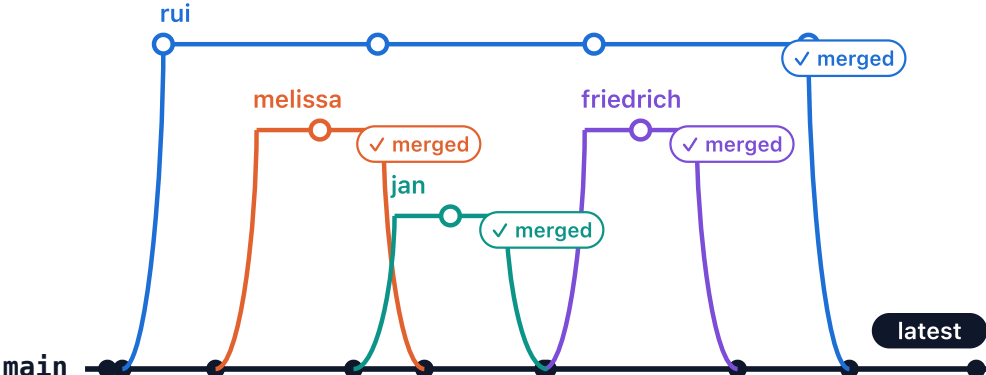


every result is one script run

```
bash
$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.00025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias -0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

VERSION CONTROL

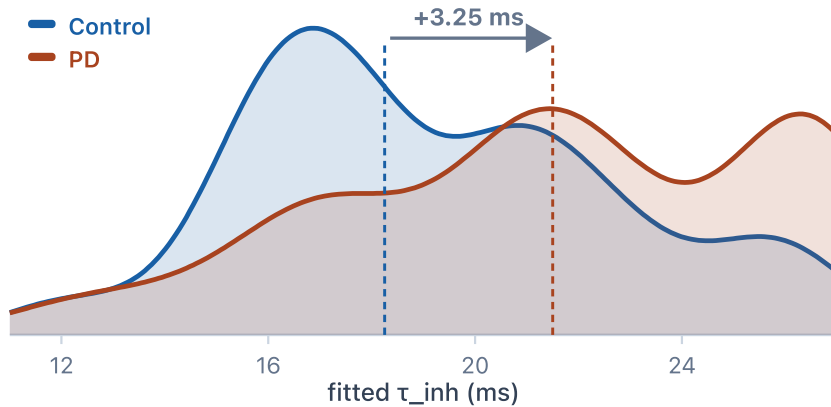
feature branches, reviewed, merged into main



50 commits · feature branches → reviewed → merged into main · fully reproducible

Does the model separate patients from controls?

THE SEPARATION fitted τ_{inh} per subject, Control vs PD

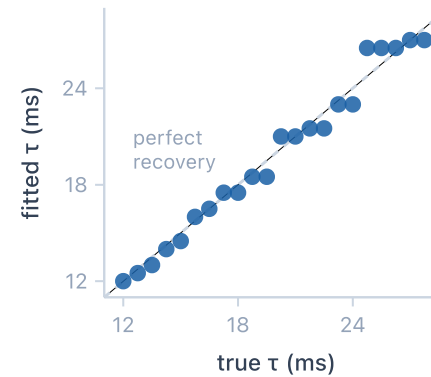


Medians differ by +3.25 ms (Control 18.25, PD 21.5 ms).

But individuals overlap: a random PD is higher only $\approx 68\%$ of the time.

IS THE DIFFERENCE REAL?

validate the fit, then compare to the effect



The fit is unbiased.

mean error -0.1 ms

typical error ± 0.5 ms

worst error 1.75 ms

The +3.25 ms difference is 6 $\times$ the fitting error:

fitting error ± 0.5 ms

group difference +3.25 ms

The fitted τ_{inh} re-expresses the measured slowing in model units. What the model adds is a **validated** fit: parameter recovery is unbiased, and the **+3.25 ms** group difference (Mann-Whitney, $p \approx 0.0002$) is 6 $\times$ the fitting error: a real shift, not an artefact.

Outlook and open challenges

CURRENT CHALLENGES

- One parameter (τ_{inh}) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted τ to clinical scores and to known inhibitory (GABAergic) changes in PD

What we took away

Rui [your biggest learning]

Jan [your biggest learning]

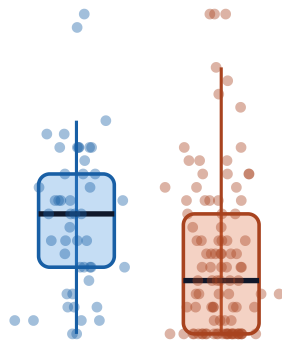
Melissa [your biggest learning]

Friedrich [your biggest learning]

Appendix: per-subject distributions

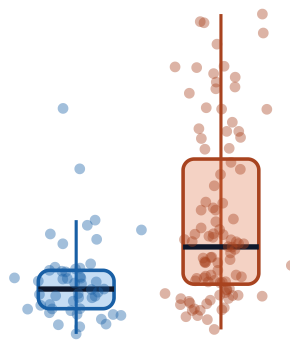
PD vs Control: per-subject EEG features

Alpha peak frequency
slower in PD



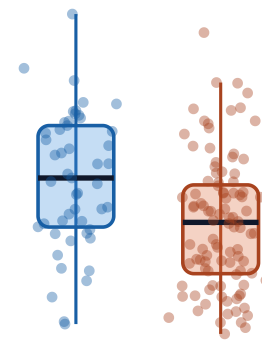
Control PD

Theta power
higher in PD



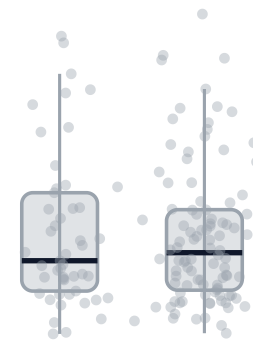
Control PD

Beta power
lower in PD



Control PD

Alpha power
unchanged



Control PD

Appendix: effect sizes

How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval

